The empirical recurrence for OEIS A236823: recovering a conjecture that is not written down

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Abstract

OEIS A236823 records "Empirical recurrence of order 84", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 177$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236823 is "Number of (n+1)X(3+1) 0..2 arrays with the maximum plus the minimum minus the lower median of every 2X2 subblock equal.". It has offset 1 and begins

$$1386, 15254, 168564, 1949236, 22695802, 269108894, 3207539588, 38599508572, \ \ldots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:36:10, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 177$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 177$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 354$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) \ = \ \sum_{i=1}^{84} c_i \, a(n-i),$$

c_1	34	c_{22}	6275038104321831719	c_{43}	−1532064723870954677241380965	c_{64}	
c_2	−103	c_{23}	67616368951230545663	c_{44}	−2794698260754004365219828925	c_{65}	
c_3	−8609	c_{24}	−129431066864674607728	c_{45}	4343288680528276372010008728	c_{66}	
c_4	86068	c_{25}	−701885751998689690459	c_{46}	5111013132962890267097372151	c_{67}	
c_5	815732	c_{26}	1842199134597224285642	c_{47}	−9975030255471260423264760749	c_{68}	
c_6	−14273082	c_{27}	5801780141253202951792	c_{48}	−7397017882009904019127916844	c_{69}	
c_7	−24567497	c_{28}	−20054507455432337004236	c_{49}	18732818479438897085795246300	c_{70}	
c_8	1251171982	c_{29}	−37601075510334374354200	c_{50}	8023879209490931262215825145	c_{71}	
c_9	−1875641980	c_{30}	173932539447048628858141	c_{51}	−28849302286856350040028381383	c_{72}	
c_{10}	−69522659097	c_{31}	182538383691400067475697	c_{52}	−5530805383617929263067292405	c_{73}	
c_{11}	249574494114	c_{32}	−1226846636654254036771934	c_{53}	36388937924928835098410575334	c_{74}	
c_{12}	2625415958883	c_{33}	−567712890836324465562644	c_{54}	221725134779149859604178876	c_{75}	
c_{13}	−14468110723020	c_{34}	7119362338017644807732667	c_{55}	−37424801669105702649176257410	c_{76}	
c_{14}	−68553067413666	c_{35}	109954859821876993999127	c_{56}	5402968858666800746190790160	c_{77}	
c_{15}	548598961086380	c_{36}	−34206718762168298395052287	c_{57}	31149570381553597701315637640	c_{78}	
c_{16}	1178093762057447	c_{37}	11871611698282821585516871	c_{58}	−8422526761908424852590529656	c_{79}	
c_{17}	−15111156938897402	c_{38}	136475735578962630940528860	c_{59}	−20758607923252364513095002272	c_{80}	
c_{18}	−9582286535795279	c_{39}	−89989965590268956276346092	c_{60}	7889504633887572729985666816	c_{81}	
c_{19}	317027396438864059	c_{40}	−452117035051680658878125212	c_{61}	10916876213474827562358616928	c_{82}	
c_{20}	−127453639005475181	c_{41}	428378600420008851928975785	c_{62}	−5245783428164741652135846976	c_{83}	
c_{21}	−5198854880988239696	c_{42}	1239808949588555451386976377	c_{63}	−4443466584131892538570682656	c_{84}	

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84+4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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